

Design and Implementation of a Purple Company E-Commerce Database

CHAPTER 1

1 Introduction

The Purple Company E-Commerce Database is designed to represent the data structure of an online beauty and personal care retail platform. Purple is one of India's leading beauty e-commerce companies, offering skincare, makeup, haircare, fragrances, wellness products, and beauty accessories through its online platform. The company manages thousands of products, customers, suppliers, and daily transactions, making an efficient database essential for smooth business operations.

This project focuses on designing a relational database that stores and manages information related to customers, products, categories, brands, orders, payments, inventory, reviews, and wishlists. These entities are connected through primary and foreign keys to maintain data consistency and integrity.

The database is implemented using Oracle SQL by applying database concepts such as normalization, constraints, relationships, and SQL operations. The system enables efficient storage, retrieval, updating, and management of business data while reducing redundancy and improving overall performance.

1.1 Problem Statement

Purple handles a large amount of customer and product information every day. Managing this data manually or without a properly structured database can lead to data duplication, inconsistencies, inventory errors, delayed order processing, and inefficient customer service.

A centralized relational database is required to organize and manage business information effectively. The system should ensure data accuracy, maintain relationships among different entities, support secure transactions, and enable efficient order and inventory management.

1.2 Objectives

database for Purple's e-commerce business.

To identify major entities such as Customers, Products, Categories, Brands, Orders, Payments, Inventory, Reviews, and Wishlist.

To establish relationships using Primary Keys and Foreign Keys.

To apply normalization techniques to reduce data redundancy.

To implement database constraints to maintain data integrity.

To perform SQL operations for inserting, updating, deleting, and retrieving records.

To support efficient order processing and inventory management.

To improve data accuracy, consistency, and reliability.

To gain practical knowledge of Oracle SQL database design.

1.3 Motive

The primary motive of this project is to design a well-structured relational database that supports the business operations of Purple. As a leading beauty e-commerce company, Purple processes thousands of customer orders and manages a wide variety of beauty products.

This project provides practical experience in applying database concepts such as entity relationship modeling, normalization, constraints, and SQL queries. It also demonstrates how a real-world e-commerce database supports efficient business operations and decision-making.

1.4 Stakeholders

The major stakeholders involved in the Purple E-Commerce Database are:

Customers

Company Management

Sales Department

Inventory Managers

Warehouse Staff

Suppliers and Brands

Finance and Payment Team

Customer Support Team

Database Administrator (DBA)

System Administrator

1.5 Business Requirements

The database should satisfy the following business requirements:

- Maintain customer registration and profile information.
- Store product details under various categories and brands.
- Manage product inventory and stock availability.
- Process customer orders efficiently.
- Record payment transactions securely.
- Track order status and delivery information.
- Allow customers to maintain wishlists.
- Store customer product reviews and ratings.
- Generate reports for management and business analysis.
- Ensure accurate inventory updates after every purchase.

1.6 Functional Requirements

The system should provide the following functions:

- Customer Registration and Login
- Product Management
- Category Management
- Brand Management
- Product Search
- Shopping Cart Management
- Wishlist Management
- Order Placement
- Payment Processing
- Inventory Management
- Review and Rating System
- Order Tracking
- Customer Profile Management
- Report Generation

1.7 Non-Functional Requirements

The database should satisfy the following quality requirements:

- High Performance
- Data Security
- Data Integrity
- Reliability
- Scalability
- Availability
- Maintainability
- Backup and Recovery Support
- User-Friendly Data Management
- Fast Query Processing
- Secure Payment Information Storage
- Compliance with Database Standards

1.8 Conclusion

The Purple Company E-Commerce Database provides a structured and efficient solution for managing the company's business operations. By organizing customer, product, order, payment, inventory, and review information within a relational database, the system improves data consistency, minimizes redundancy, and enhances operational efficiency.

The implementation of Oracle SQL concepts such as normalization, constraints, primary keys, foreign keys, and SQL operations ensures reliable and accurate data management. This project demonstrates how a well-designed relational database supports the daily activities of a modern beauty e-commerce company while providing a scalable and secure foundation for future business growth.